

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0349

Barling 345 kV
May 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0349. The Interconnection Customer proposes a 689.3 MW natural gas combined cycle facility interconnecting at the Barling 345 kV substation in the Maryland area. The request was received on June 27, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 167 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,784,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Connell-Chiquita 345kV	Tombstone Substation-Marcus Hook Energy Lp 500kV	120.9	1,196	5.6
Connell-Tombstone Substation 345kV	Tombstone-Marcus Hook Energy Lp Substation 500kV	116.7	1,328	5.8
Chiquita-Irvine 345kV L	Tombstone-Marcus Hook Energy Lp 500kV	108.1	1,176	5.3
Sharyland-Bixby Hill 345kV	Tombstone-Bixby Hill 500kV L	102.0	2,180	13.2
Tombstone-Sharyland 345kV	Bixby Hill-Tombstone 500kV	101.1	2,247	12.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Connell-Chiquita 345kV	Rebuild line with higher-capacity conductor	\$28,675,000	17% (\$4,894,000)
Connell-Tombstone 345kV	Rebuild line with higher-capacity conductor	\$23,762,000	18% (\$4,172,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chiquita-Irvine 345kV L	Reconductor line	\$13,537,000	17% (\$2,298,000)
Sharyland-Bixby Hill 345kV	Reconductor line	\$21,825,000	9% (\$1,892,000)
Tombstone-Sharyland 345kV line	Reconductor line	\$18,856,000	8% (\$1,528,000)
	Total allocated to project		\$14,784,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.